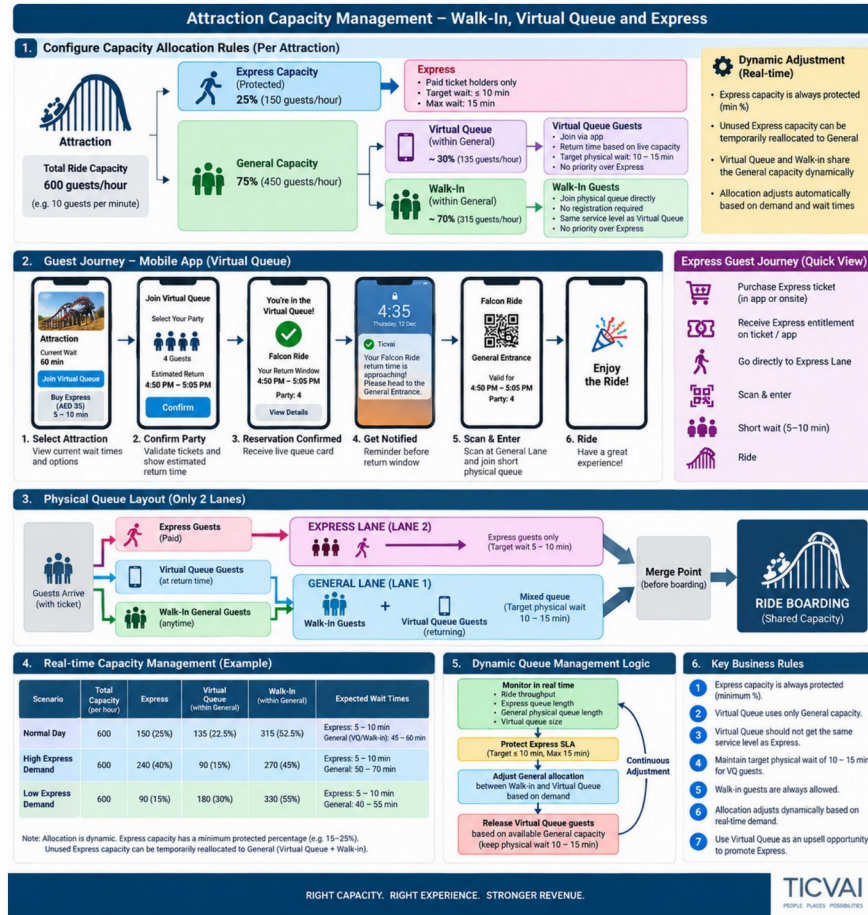


TICVAI

Virtual Queue Management

Detailed Functional Explanation, Use Cases & Operational Examples



Functional reference for Business, Operations, Product, UI/UX and Development teams

Document Purpose

This document explains the four TICVAI Virtual Queue boards and the function of each major section. It translates the visual concepts into practical operating behavior for business stakeholders, operations teams, product owners, UI/UX designers and developers.

The explanations are grounded in the supplied board designs. Operational examples are illustrative scenarios showing how the configured concepts could be applied.

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1. Virtual Queue Configuration for Rides - Admin Screens



Board objective: Configure attractions, queue types, capacity, operating rules, schedules, reservations and guest notifications.

1.1 Attractions List

Purpose: Provides the master list of rides and attractions that can participate in queue management.

How it works: Users can search attractions, view type/status, add an attraction, or open an attraction for configuration.

Use case: An administrator needs to activate Virtual Queue for a newly opened ride.

Example: Falcon Coaster is selected while Sky Tower remains inactive; only Falcon Coaster's queue configuration is changed.

Operational benefit: Creates one controlled configuration point for each attraction.

1.2 Ride Details & Queue Settings

Purpose: Defines the attraction and which queue experiences are enabled.

How it works: Virtual Queue, Express and physical walk-in can be enabled. Queue mode can be Hybrid, Virtual Queue Only or Physical Queue Only. The example shows 600 guests/hour or 10 guests/minute.

Use case: Falcon Coaster must support Walk-In, VQ return guests and paid Express guests.

Example: Hybrid mode is selected and both Virtual Queue and Express are enabled.

Operational benefit: Supports different queue models per attraction.

1.3 Capacity Allocation

Purpose: Defines how throughput is shared between General, Virtual Queue and Express.

How it works: The board shows 70% General, 20% VQ and 10% protected Express. At 600 guests/hour this equals 420, 120 and 60 guests/hour. Unused Express capacity can be temporarily reallocated to General.

Use case: General demand is high while Express arrivals are below allocation.

Example: Unused Express capacity can temporarily support General/VQ demand while the Express protection rule remains in place.

Operational benefit: Improves utilization while protecting the premium queue.

1.4 Virtual Queue Rules

Purpose: Controls the service promise and eligibility rules for VQ guests.

How it works: Settings include target physical wait after return, return-window duration, early/late tolerance, max party size, max active reservations and no-show handling.

Use case: A family returns slightly late for its reserved window.

Example: The configured return window and late-arrival tolerance determine whether entry is still accepted; active-booking limits protect slot inventory.

Operational benefit: Creates predictable guest rules and reduces misuse.

1.5 Express Queue Rules

Purpose: Protects the expected premium Express experience.

How it works: Settings include target wait SLA, maximum wait, minimum/default/maximum capacity, reallocation of unused capacity and merge-point priority. The board shows target ≤ 10 minutes and maximum 15 minutes.

Use case: Express wait rises toward the maximum threshold.

Example: Capacity can be increased within configured limits and Express can retain merge priority until service recovers.

Operational benefit: Links Express entitlement to a measurable premium service level.

1.6 Operating Schedule & Capacity

Purpose: Varies capacity and allocation by time period or special event.

How it works: Time slots can have different hourly capacity and General/VQ/Express percentages. The sample ranges from 400 to 800 guests/hour.

Use case: The ride has fewer trains in the morning and more capacity during the afternoon.

Example: Morning is configured at 400/hour and the peak period at 800/hour with a different allocation split.

Operational benefit: Aligns digital inventory with actual ride capability.

1.7 Live Queue Monitor

Purpose: Shows current queue conditions and throughput.

How it works: Displays total waiting, counts by queue type, estimated waits and ride throughput versus target.

Use case: A supervisor needs to know why Falcon Coaster is congested.

Example: The board shows 312 waiting, separate queue waits and throughput of 580 against a 600/hour target.

Operational benefit: Supports real-time operational decisions.

1.8 Virtual Queue Reservations

Purpose: Lists individual VQ reservations for operational management.

How it works: Reservations show guest, party size, return window and status such as Waiting, Notified or Arrived, with filtering and actions.

Use case: A guest asks staff to confirm a return booking.

Example: The attendant searches the reservation, verifies the window/status and handles the case using permitted actions.

Operational benefit: Supports guest service and exception management.

1.9 Notifications & Messages

Purpose: Configures automated guest communications.

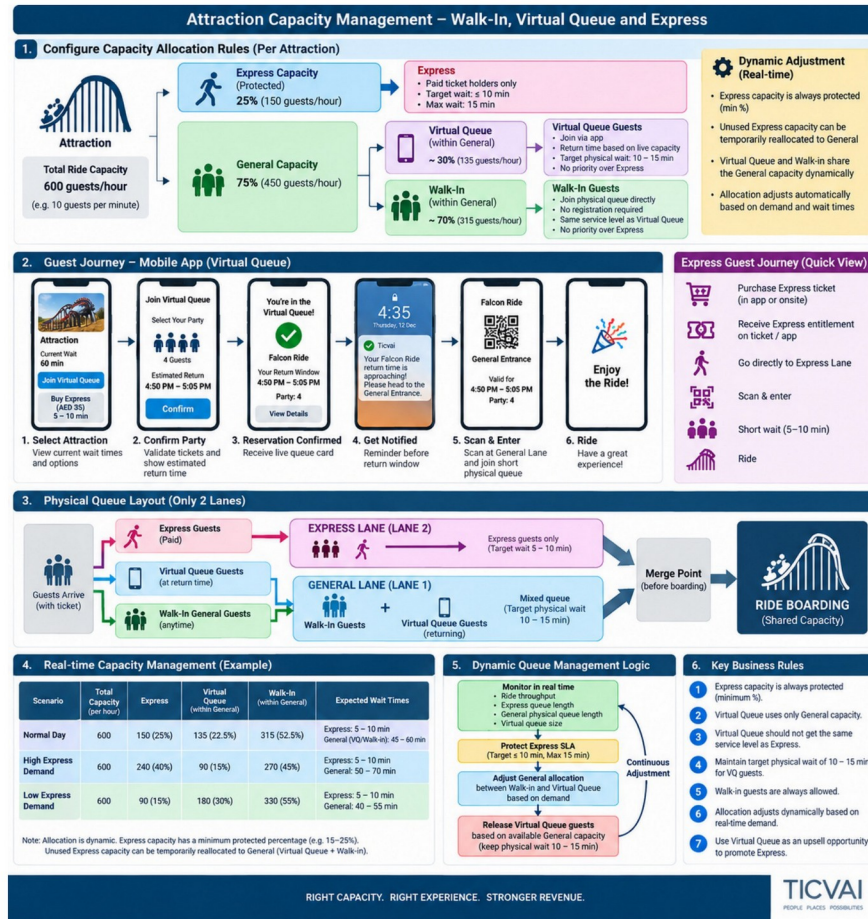
How it works: Triggers include booking confirmation, return reminder, It's Your Turn, late arrival, cancellation, ride closure/delay and marketing/upsell.

Use case: A guest has a 10:30 return window.

Example: A reminder can be sent 15 minutes before, followed by an It's Your Turn notification at return time; disruption messages can be sent if needed.

Operational benefit: Reduces missed windows and keeps guests informed.

2. Attraction Capacity Management - Walk-In, Virtual Queue and Express



Board objective: Explain how three guest experiences share attraction capacity and two physical lanes.

2.1 Capacity Allocation Rules

Purpose: Translates total ride capacity into protected Express and shared General capacity.

How it works: The illustration uses 600 guests/hour, with 25% protected Express (150/hour) and 75% General (450/hour). Virtual Queue and Walk-In are managed within General capacity.

Use case: The attraction has only two physical lanes but offers three access products.

Example: Express uses its dedicated lane; returning VQ and Walk-In guests share the General lane while capacity is managed digitally.

Operational benefit: Supports three guest products without requiring three physical lanes.

2.2 Virtual Queue Mobile Journey

Purpose: Shows the guest journey from joining VQ to boarding.

How it works: Guest checks current wait, selects party, confirms VQ, receives a return window, gets notified, scans at General Entrance and rides.

Use case: A family of four does not want to stand in a 60-minute queue.

Example: They reserve a 4:50-5:05 PM return window, enjoy the park, receive a reminder and return to scan into the General lane.

Operational benefit: Converts standing time into usable park time.

2.3 Express Guest Journey

Purpose: Defines the premium fast-access journey.

How it works: Guest buys/receives Express, goes directly to the Express lane, scans and enters, with a short target wait shown as about 5-10 minutes.

Use case: A guest buys Express because the General wait is long.

Example: The entitlement is validated at the Express entrance and the guest bypasses VQ reservation.

Operational benefit: Provides a differentiated premium experience.

2.4 Physical Queue Layout - Only 2 Lanes

Purpose: Explains physical handling of Walk-In, VQ and Express.

How it works: Express uses Lane 2. Walk-In and returning VQ guests use General Lane 1. Both streams meet at a merge point before boarding.

Use case: There is no space to build a third VQ lane.

Example: VQ guests are admitted to Lane 1 only during valid return windows, while Express remains in Lane 2.

Operational benefit: Minimizes infrastructure while preserving Express separation.

2.5 Real-Time Capacity Management

Purpose: Shows allocation changes under different demand conditions.

How it works: Normal, High Express Demand and Low Express Demand scenarios redistribute the 600/hour capacity while maintaining Express protection and General sharing.

Use case: A large group creates a spike in Express arrivals.

Example: Express allocation can rise while VQ release and Walk-In throughput are adjusted to keep the premium wait under control.

Operational benefit: Allows allocation to follow real demand.

2.6 Dynamic Queue Management Logic

Purpose: Defines the continuous feedback loop for queue control.

How it works: Monitor throughput and queue lengths, protect Express SLA, adjust Walk-In/VQ allocation and release VQ guests according to available General capacity while maintaining target physical wait.

Use case: VQ return guests begin arriving faster than boarding capacity.

Example: Future VQ slot release can be reduced until the physical General lane returns toward the 10-15 minute VQ target.

Operational benefit: Prevents Virtual Queue from becoming another long physical queue.

2.7 Key Business Rules

Purpose: Summarizes the governing principles for capacity decisions.

How it works: Express is protected; VQ uses General capacity; VQ does not receive the same service level as Express; target VQ physical wait is 10-15 minutes; Walk-In remains allowed; allocation is dynamic; VQ can support Express upsell.

Use case: All three demand types are high simultaneously.

Example: The system preserves Express protection and manages VQ release within General capacity rather than eliminating the VQ wait by consuming premium capacity.

Operational benefit: Maintains fairness, product differentiation and operational consistency.

3. Operations Control Room



Board objective: Provide a live command center for queues, throughput, capacity, alerts, incidents and guest flow.

3.1 Park-Level KPI Summary

Purpose: Shows the overall operating health of the park.

How it works: Cards show guests in park, average wait, Express guests, VQ guests, ride utilization and guest satisfaction with comparison indicators.

Use case: A duty manager starts the afternoon shift.

Example: High utilization plus rising waits directs attention to attractions and alerts needing intervention.

Operational benefit: Enables rapid park-level prioritization.

3.2 Attractions Overview

Purpose: Compares queue conditions and status across rides.

How it works: The list shows General, VQ and Express waits plus statuses such as Open, High Load or Delayed.

Use case: Operations must identify which attraction needs attention first.

Example: A delayed Haunted Mansion or high-load River Rapids can be prioritized over stable attractions.

Operational benefit: Provides one portfolio view of attraction pressure.

3.3 Selected Attraction Live Overview

Purpose: Shows detailed live metrics for the selected ride.

How it works: General, VQ and Express waits and throughput versus target are displayed, with tabs for queue details, capacity, operations, incidents and guest flow.

Use case: Falcon Coaster's General wait increases to 45 minutes.

Example: The supervisor sees 580/hour throughput against 600 target and compares all three queue waits.

Operational benefit: Helps identify whether the issue is capacity, demand or allocation.

3.4 Queue Segment Detail

Purpose: Breaks demand into General, VQ and Express pools.

How it works: Tiles show waiting/queued guests, estimated wait/return time and occupancy against allocation.

Use case: VQ is heavily occupied while Express is lightly used.

Example: Operations can compare General 320, VQ 214 and Express 42 in the example before deciding whether to adjust allocation.

Operational benefit: Makes queue decisions evidence-based.

3.5 Real-Time Capacity Allocation

Purpose: Visualizes the current percentage/hourly allocation by queue type.

How it works: A 600 guests/hour capacity split is shown with an Adjust Allocation control.

Use case: Express demand rises above the desired service level.

Example: An authorized operator can change allocation within business rules to stabilize service.

Operational benefit: Allows controlled intervention without redesigning the ride setup.

3.6 Queue Trend - Last 4 Hours

Purpose: Shows how waits evolve over time.

How it works: Trend lines compare General, VQ and Express wait behavior.

Use case: Current General wait is 45 minutes, but the manager needs to know whether it is improving.

Example: A falling trend may indicate that intervention is already working, avoiding unnecessary changes.

Operational benefit: Prevents decisions based on a single snapshot.

3.7 Guest Arrivals vs Throughput

Purpose: Compares incoming demand with ride processing rate.

How it works: The chart plots arrivals against throughput.

Use case: Queue length rises even though the ride has no incident.

Example: If arrivals consistently exceed throughput, operations can reduce VQ release, redirect guests or adjust allocation.

Operational benefit: Explains the operational cause of queue growth.

3.8 Live Alerts

Purpose: Highlights conditions requiring attention or action.

How it works: Examples include high queue length, increasing Express queue, throughput below target, high VQ no-show rate and weather advisory.

Use case: Express wait reaches 12 minutes.

Example: The alert flags the risk and provides an action path for investigation or capacity adjustment.

Operational benefit: Reduces response time to emerging problems.

3.9 Quick Actions

Purpose: Provides immediate controls for common interventions.

How it works: Actions include Pause Virtual Queue, Adjust Capacity Allocation, Send Guest Notification, Manage Reservations, Open/Close Ride and Report Incident.

Use case: A ride temporarily stops for inspection.

Example: The operator pauses new VQ bookings, notifies guests and records the incident.

Operational benefit: Centralizes monitoring and intervention.

3.10 Live Park Map

Purpose: Adds spatial context to queue and incident data.

How it works: The map shows attraction locations and current operational context.

Use case: Several nearby attractions become busy together.

Example: Operations recognizes a congestion zone and redirects guests toward less busy areas.

Operational benefit: Supports area-level crowd management.

4. AI Guest Flow Optimization, Analytics & Engagement



Board objective: Predict future pressure, recommend interventions, optimize capacity and influence guest flow.

4.1 KPI & Prediction Summary

Purpose: Combines current performance with forward-looking indicators.

How it works: Shows guests in park, average wait, predicted peak, ride utilization, satisfaction and app engagement.

Use case: Current waits are acceptable but management wants to know what happens later.

Example: A predicted peak of about 18,500 guests between 16:00-18:00 indicates preventive action may be needed.

Operational benefit: Adds future context to current KPIs.

4.2 Live Park Map & Guest Density

Purpose: Shows where guests and congestion are concentrated.

How it works: Views include guest density, wait times and predicted congestion with attraction markers.

Use case: Falcon Coaster is predicted to become highly congested from 16:00-18:00.

Example: Operations prepares staffing, queue controls or redirection before the peak occurs.

Operational benefit: Supports proactive spatial crowd management.

4.3 AI Demand Forecast

Purpose: Forecasts guest demand across the day and future periods.

How it works: Actual guest levels are compared with an AI forecast and predicted peak window.

Use case: Late-afternoon arrivals are expected to rise sharply.

Example: The forecast gives operations advance time to adjust capacity or communications.

Operational benefit: Enables planning before demand reaches the queues.

4.4 AI Recommendations - Real Time

Purpose: Turns conditions into prioritized suggested actions.

How it works: Examples include increasing Express allocation, restricting VQ slots, promoting alternatives, opening another train and proactive weather messaging, with priority and action buttons.

Use case: Falcon Coaster Express wait reaches 12 minutes.

Example: AI recommends increasing Express capacity from 10% to 20% for the next two hours; an operator reviews and applies it.

Operational benefit: Converts analytics into actionable decisions while retaining human control.

4.5 Attraction Wait-Time Prediction

Purpose: Shows current and predicted future waits per attraction.

How it works: The table includes current wait, 60-minute prediction, two-hour prediction, trend and status.

Use case: Haunted Mansion is currently 55 minutes and predicted to rise.

Example: Operations can redirect demand or investigate throughput before the queue becomes critical.

Operational benefit: Supports preventive rather than reactive queue management.

4.6 Guest Flow Optimization

Purpose: Recommends allocation and routing changes.

How it works: The board shows capacity splits and AI-suggested adjustments, with tabs for capacity allocation, guest routing and scenario simulation.

Use case: One attraction is overloaded while another has spare capacity.

Example: The system can recommend changing allocation or promoting the less busy attraction.

Operational benefit: Balances demand across available park capacity.

4.7 Guest Engagement

Purpose: Uses app campaigns, onsite messaging and personalization to influence behavior.

How it works: Examples include River Rapids discount, crowd messaging, Express upgrade and family-zone promotion.

Use case: Aqua Splash is crowded while River Rapids has spare capacity.

Example: Guests near Aqua Splash receive a targeted River Rapids promotion.

Operational benefit: Links operational optimization with guest engagement and revenue opportunities.

4.8 Key Insights - AI Analytics

Purpose: Summarizes measurable operational and commercial outcomes.

How it works: The board highlights Express revenue, average queue length, ride throughput and satisfaction changes.

Use case: Management evaluates whether the strategy is working.

Example: Lower average queue length with higher throughput and Express revenue indicates positive operational/commercial impact.

Operational benefit: Provides management-level evidence of results.

4.9 What-If Scenario Simulation

Purpose: Tests a proposed change before operational application.

How it works: The sample 'Add 1 Train' scenario estimates changes to wait time, throughput and revenue.

Use case: Operations considers opening an additional train on Falcon Coaster.

Example: The board estimates wait reducing 45 to 28 minutes, throughput increasing 600 to 750/hour and revenue +12%. These are scenario outputs shown by the design, not guaranteed outcomes.

Operational benefit: Supports informed resource decisions.

4.10 Alerts & Events

Purpose: Shows a chronological feed of predicted and actual optimization events.

How it works: Examples include predicted congestion, Express wait above target, throughput below target, high no-show rate and weather advisory.

Use case: Weather may affect an outdoor ride later.

Example: Operations receives advance warning and prepares guest communication or capacity changes.

Operational benefit: Combines predictive and operational events in one timeline.

5. End-to-End Operational Scenario

Falcon Coaster - Afternoon Peak

- 1. Base configuration** - Hybrid operation is enabled for Walk-In, Virtual Queue and Express, with total capacity of 600 guests/hour.
- 2. Capacity allocation** - Express receives protected capacity while General capacity supports Walk-In and VQ according to configured rules and schedule.
- 3. Demand increases** - General wait rises and the VQ pool grows. The Control Room compares arrivals, throughput and queue trends.
- 4. Predictive warning** - The AI board forecasts heavier demand and identifies Falcon Coaster as a likely congestion point.
- 5. Recommended action** - TICVAI can recommend restricting new VQ slots, changing Express allocation if its SLA is at risk, promoting alternative attractions, or adding operational capacity where available.
- 6. Guest communication** - Existing VQ guests continue receiving return-window messages; new guests may see reduced VQ availability or alternative recommendations.
- 7. Physical arrival** - Returning VQ guests scan into the General lane; Express guests use the protected Express lane; both streams merge before boarding.
- 8. Continuous adjustment** - Throughput and waits are monitored and VQ availability/allocation can be restored as conditions improve.
- 9. Outcome measurement** - Operations reviews queue length, throughput, satisfaction and commercial indicators to assess the intervention.

6. Cross-Board Functional Relationship

Board	Primary Role	Main User	Typical Decision
Configuration	Define rules and attraction behavior	Administrator / Operations Manager	How should this ride's queues and capacity operate?
Capacity Management	Define how guest types share capacity and lanes	Operations / Product	How do Walk-In, VQ and Express coexist?
Operations Control Room	Monitor and intervene in real time	Duty Manager / Control Room	What is happening now and what action is required?
AI Optimization	Predict, recommend and optimize	Operations / Management	What is likely to happen next and what should we change?

Functional Design Principle

The four boards form one operating cycle: Configure -> Allocate -> Monitor -> Predict/Optimize. Configuration establishes rules; capacity management determines how guest products share throughput; the Control Room measures actual conditions; and the AI layer predicts future pressure and recommends actions.